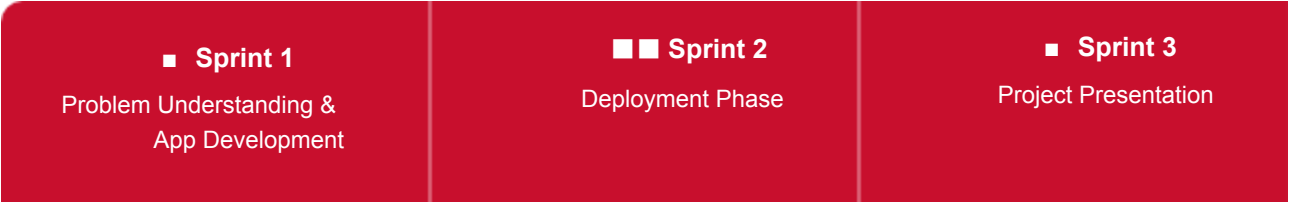




Generative AI Project Execution Guide

Innomatics Research Labs

A structured, industry-grade workflow for building, deploying & presenting
Generative AI applications using Prompt Templates, LLMs & Output Parsers.



■ Overview

This guide provides a common structured workflow for all Generative AI projects. Each project is executed across **3 Sprints**, ensuring industry-level development practices from inception to presentation.

Technology	Purpose
Prompt Templates	Structured, reusable input formatting for LLMs
Chat Models (LLM)	Core AI inference and natural language generation
Pydantic / JSON Parsers	Structured, validated output from model responses



■ Sprint 1 — Problem Understanding & Application Development

Objective: Understand the problem statement clearly and build a working application using Generative AI.

Phase 1 · Problem Understanding

- Read the problem statement thoroughly
- Identify the input type (text, paragraph, query, etc.)
- Define the expected output (structured format using Pydantic/JSON)
- Clarify the business use case
- Define what exactly the chatbot should do — and what it should NOT include

Phase 2 · Design the Solution

Design the full system flow:

User Input	Prompt Template	LLM	Output Parser	Structured Output
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Deliverables:

- Flow diagram
- Prompt template draft
- Output schema definition

Define: Prompt structure & Output schema (Pydantic model or JSON schema)

Phase 3 · Application Development

Setup project environment and implement the following components:

Component	File
Prompt Template	prompt.py
LLM Call	model.py
Output Parser (Pydantic/JSON)	parser.py
Entry Point	main.py

Interface: Build a simple CLI, Streamlit, or Basic Web UI to interact with the application.



Phase 4 · Testing

- Test with multiple inputs
- Validate structured output
- Handle edge cases

■ ■ Sprint 2 — Deployment Phase

Objective: Deploy the application so it can be accessed publicly.

Phase 1 · Deployment Preparation

- Refactor code if needed
- Add **requirements.txt** for dependencies
- Add **.env** for environment variable handling
- Ensure API keys are secured — no hardcoding

Phase 2 · Deployment

Suggested Platforms:

Streamlit Cloud	Hugging Face Spaces

Ensure: App runs without errors, UI is responsive, LLM calls are working.

Phase 3 · Testing & Validation

- Test the deployed app
- Check latency
- Verify output correctness
- Validate error handling



■ Sprint 3 — Project Presentation

Objective: Present your project professionally with clarity and confidence.

Phase 1 · Documentation

1	Problem Statement
2	Objective
3	Approach
4	Architecture
5	Technologies Used
6	Prompt Design
7	Output Schema
8	Challenges & Solutions
9	Future Improvements

Prepare a structured report document containing

Phase 2 · Presentation Preparation

Create a presentation (PPT) including:

- Title Slide
- Problem Statement
- Solution Overview
- Demo Screenshots
- Architecture Diagram
- Key Learnings

Phase 3 · Live Demo

Show:

- Working application
- Example inputs & outputs

Explain:

- How the prompt works
- How the output parser structures data



Phase 4 - Q&A Viva

Be prepared to answer questions on your design decisions, implementation details, challenges faced, and potential future improvements to the application.

■ Final Submission Checklist

■ GitHub Repository	Source code with clean modular structure
■ Deployed Application Link	Publicly accessible, fully functional app
■ Project Report (PDF)	Comprehensive structured document
■ Presentation Slides	Professional PPT covering all key aspects
■ Demo Ready	Live working demo with sample inputs & outputs

Best of luck with your Generative AI project! | Innomatics Research Labs, Hyderabad

